

25 January 2023

Rare Earth Mineral Resource Drilling Underway at Circle Valley

- **Drilling at Circle Valley (MEK 100%) is underway targeting an area of ~15km²** where shallow high-grade rare earth mineralisation grading up to 6,894ppm TREO was intersected in 2022.
- Infill drilling will be completed during the March 2023 quarter with **delivery of an initial Mineral Resource targeted for the June 2023 quarter.**
- **Strike extension drilling for gold** at Anomaly A, where **23m @ 5.09g/t Au** (22CVRC001) was intersected in 2022, will also be completed.

Commenting on the start of infill and extensional drilling at Circle Valley, Meeka's Managing Director Tim Davidson said: "Results received in 2022 show a shallowing cover profile to the northwest at Circle Valley, corresponding with a +1,000ppm high-grade component of the rare earth mineralisation, rich in NdPr magnet rare earth elements. This shallow high-grade mineralisation trends northwest into an undrilled part of Circle Valley, which will be the focus for Mineral Resource drilling now underway. This drilling will enable delivery of an initial rare earth Mineral Resource during the June 2023 quarter.

Confidence to advance our rare earth projects to the Mineral Resource development stage is bolstered by the positive results from our ongoing metallurgical test program. This program shows that the mineralisation can be upgraded using low cost and simple beneficiation processes prior to metal extraction through leaching. The development of a Mineral Resource, in conjunction with the ongoing metallurgical test work, will support future economic studies.

Finally, we will commence extensional drilling at Anomaly A in the coming weeks, where broad zones of high-grade gold were intersected during 2022.

Drilling results are expected throughout February and March 2023."

Meeka Metals Limited ("**Meeka**" or "**the Company**") is pleased to announce that infill drilling for rare earths at Circle Valley is underway. The 2023 infill program will focus on the north-western zone where high TREO grades, up to 6,894ppm, are coincident with shallow cover. The mineralisation also persistently demonstrates a high proportion of the grade, up to ~30% of the TREO grade, as valuable NdPr magnet rare earths.

The Company will also be looking to build on the high-grade gold results at Anomaly A. Assay results will be ongoing during the quarter. Delivery of an initial rare earth Mineral Resource is targeted for the June 2023 quarter.



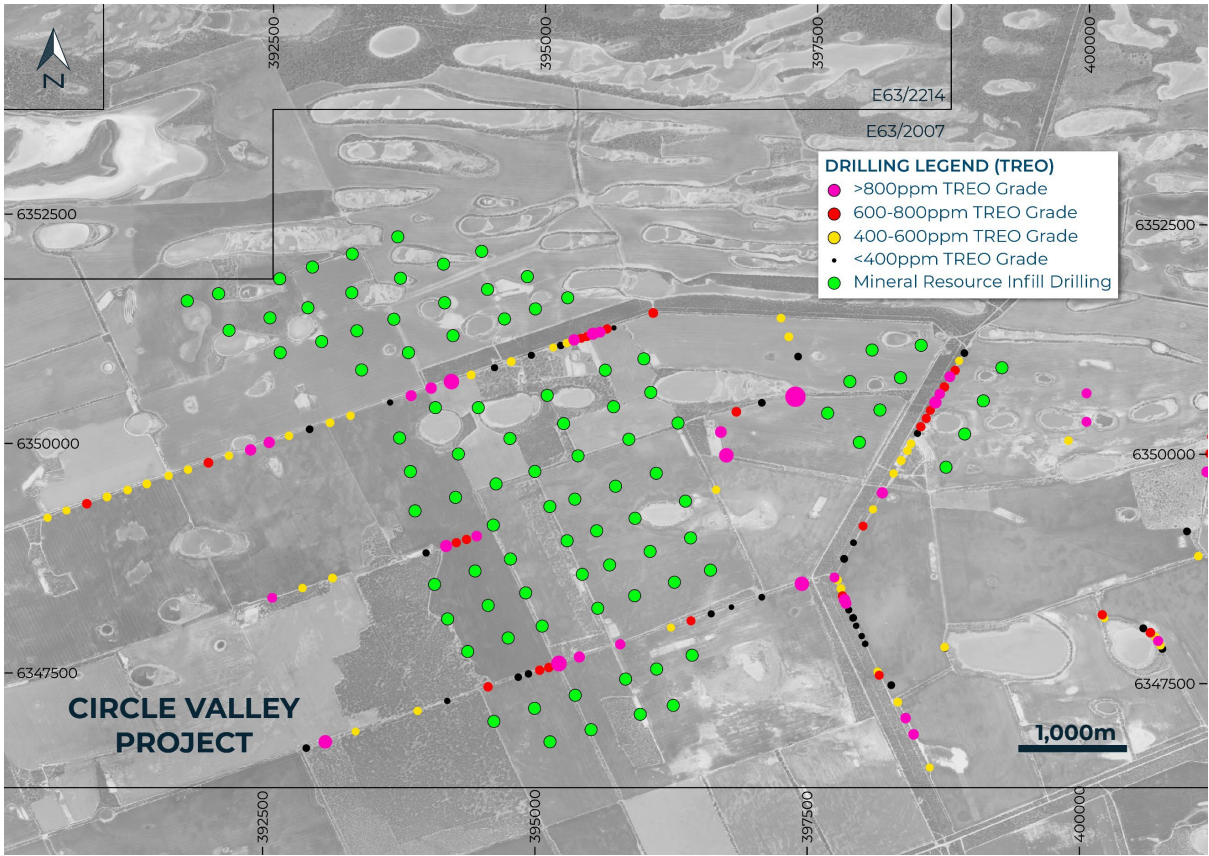


Figure 1: Plan view showing north-western zone (~15km²) of Circle Valley where rare earth Mineral Resource drilling (green dots) is underway.

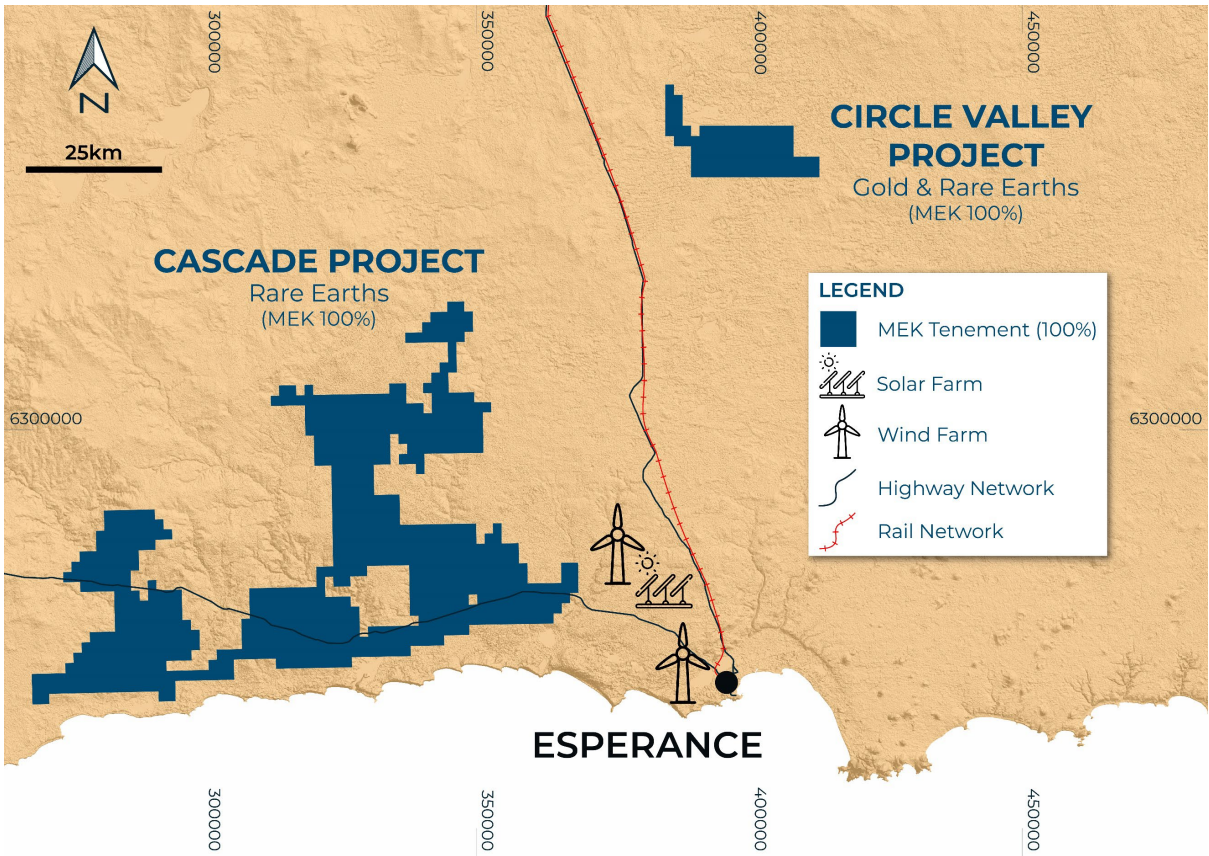


Figure 2: Plan showing the Company's 100% owned Albany-Fraser projects.

ABOUT RARE EARTH ELEMENTS

Rare earths are used in glass and ceramics, phosphors, medical imaging, communication technology, the automotive industry, electric vehicles and in renewable energy generation. The unique chemical and physical properties of rare earths have positioned them as a critical material across a number of rapidly evolving markets and industrial applications. Of particular importance are magnet rare earth elements, neodymium and praseodymium, used in the manufacture of powerful permanent magnets for electric motors and turbines.

Key global megatrends driving strong and diversified demand for Neodymium-Praseodymium oxides include:



Low carbon energy transition – electric drive motors and turbines.



Military application – guidance and control systems.



Communications technology.



Sustainable resource security – increasing scarcity of and global competition for resources.

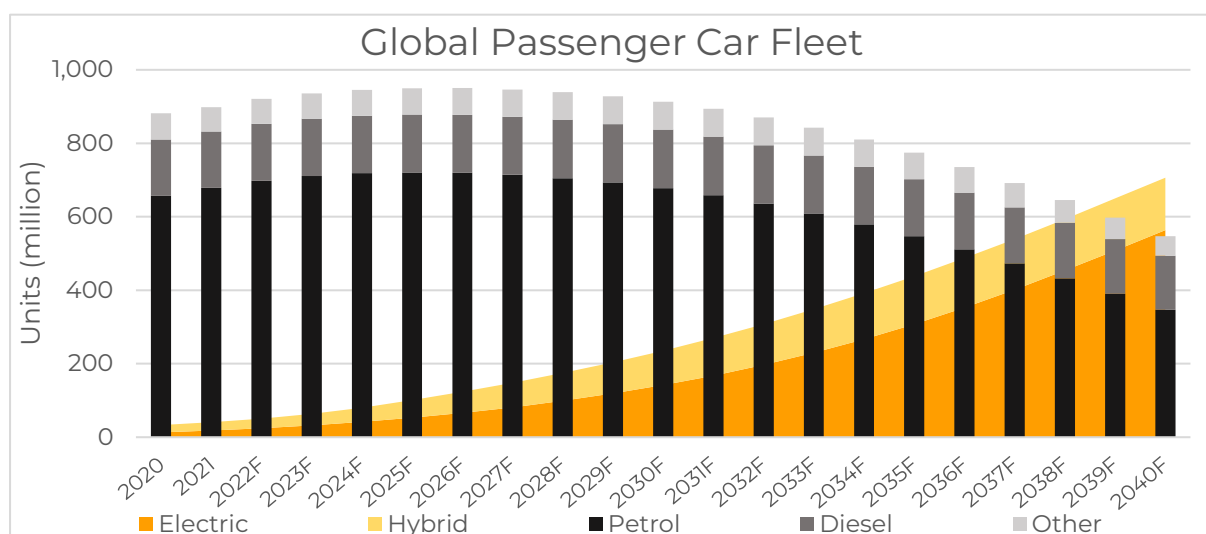


Supply chain security – against a backdrop of heightened geopolitical tension and push to diversify supply away from China.

KEY DEMAND DRIVERS FOR RARE EARTH ELEMENTS

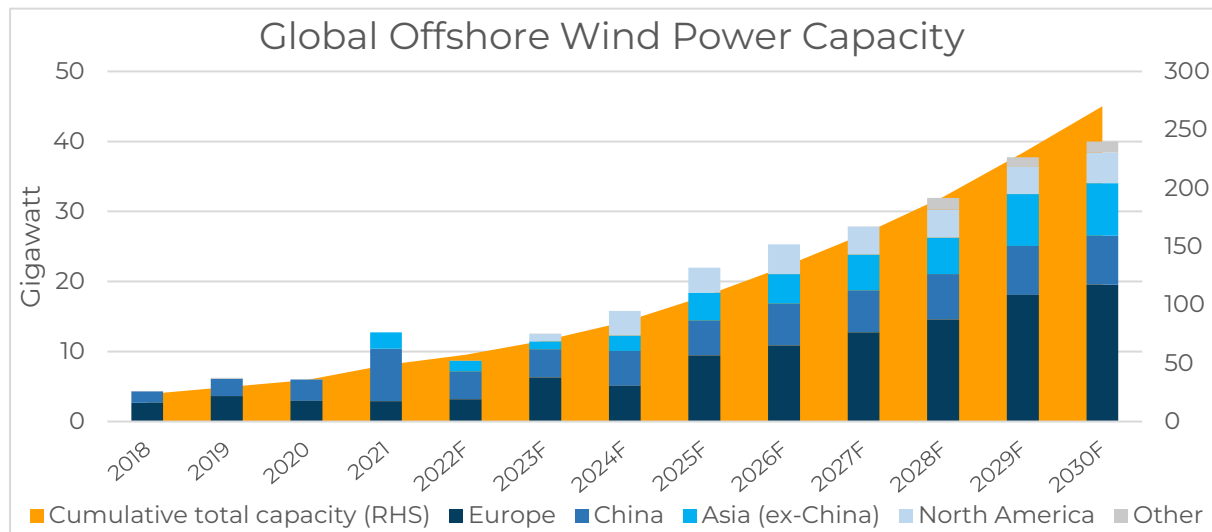
The public and private sector push toward a low carbon economy is driving increased penetration of electric vehicles (EV) and use of renewable technologies for energy generation. These megatrends drive growing demand for permanent magnets and are forecast to be the primary driver of growth in rare earth demand over next 10 years.

Global EV sales are forecast to grow at 20% CAGR to 2026 (20 million units/year). By 2040 there are forecast to be more EV's on the road than hydrocarbon powered passenger vehicles. Each EV uses 2-5kg of rare earth magnets.¹



¹ Argus, "Rare Earth Analytics", Report, April 2022.

Installed wind turbine generating capacity is forecast to grow at 25% CAGR to 2030. Each direct-drive turbine uses 650kg of rare earth magnets per megawatt of generation capacity.²



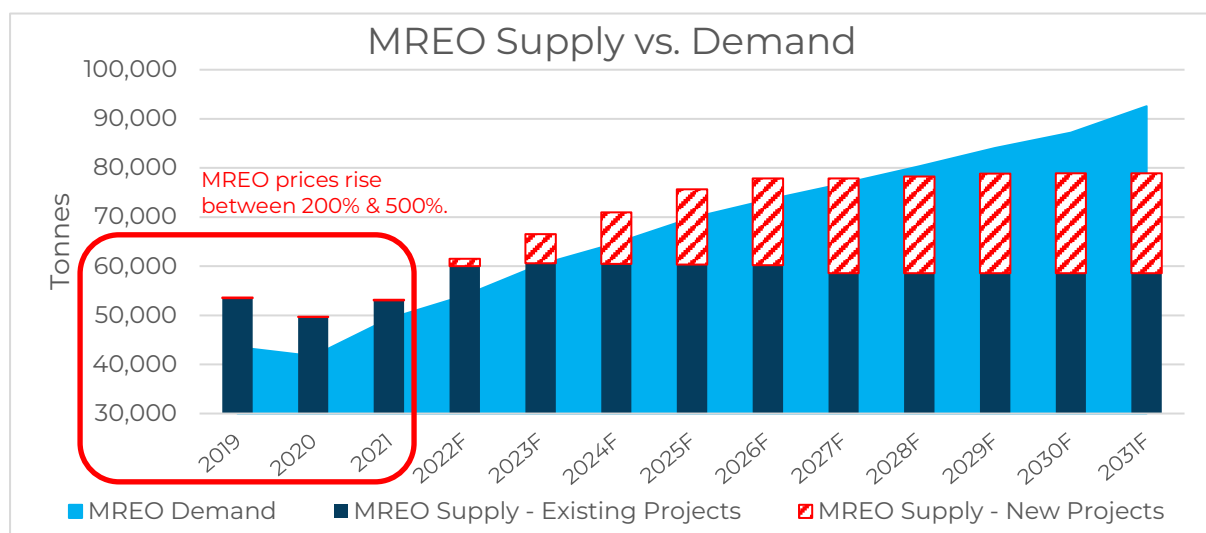
THE OPPORTUNITY – GROWING DEMAND OUTPACES SUPPLY

Global demand for magnet rare earth elements neodymium, praseodymium, dysprosium and terbium is expected to grow faster than demand for all other rare earth elements, challenging the ability of the supply-side to keep up.

Market analysts forecast a supply deficit in magnet rare earth oxide (MREO) of between 15% and 37%, within the next 5 years due to tight supply from current producers and a lack of new production coming online.³

Key points:

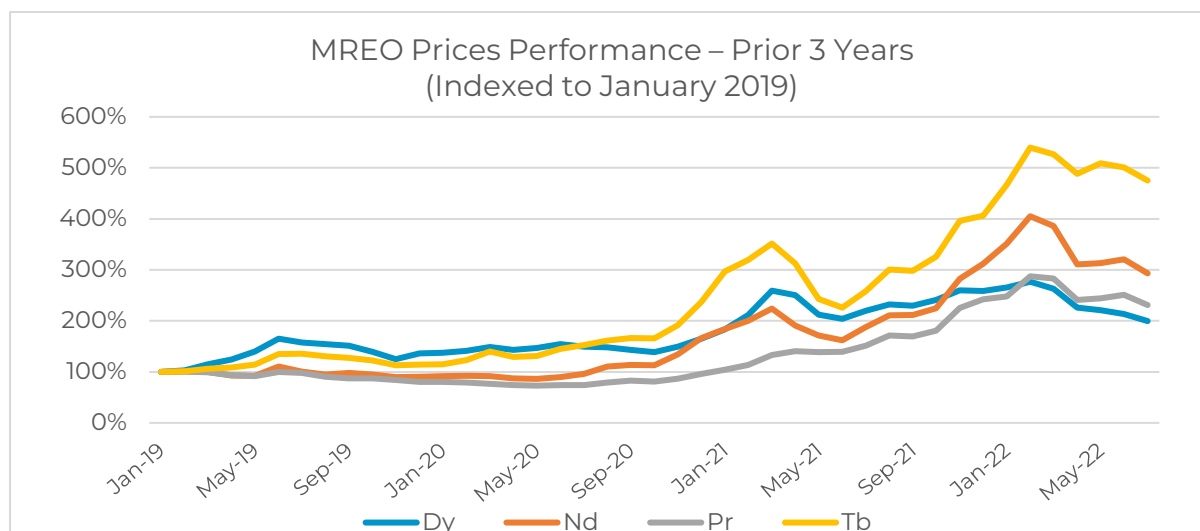
- MREO supply deficit of 37% forecast by 2031 if no new supply comes online.
- MREO supply deficit of 15% forecast by 2031 if all new sources of supply are developed and produce as forecast.



² Argus, "Rare Earth Analytics", Report, April 2022.

³ Argus, "Rare Earth Analytics", Report, April 2022.

To understand potential impact of supply shortfalls on MREO pricing, the preceding 3 years (2019 through 2021) provides a good guide. While markets were in a state of balance, MREO prices appreciated between 200% and 500%.⁴



ABOUT CLAY HOSTED RARE EARTH DEPOSITS

Clay hosted rare earth deposits often enjoy significant project and cost advantages compared to hard rock deposits, with lower cost bulk mining and a simple process flow sheet. Clay deposits do not require the higher cost comminution and beneficiation processes that hard rock deposits require, resulting in lower capital intensity and lower operating cost to produce a refined product. The generally higher proportion of magnet rare earth elements (neodymium-praseodymium) in clay deposits also results in a high value product. Additionally, clay deposits may not produce the deleterious tailings waste.

Criteria	Clay Hosted REE	Hard Rock Hosted REE
Mineralisation 	Elevated MREO.	Can be either LREO or HREO dominant mineralisation.
Resource Definition 	- Rapid, shallow, drilling into clay. - Lower cost.	- Slow, deeper, drilling into hard rock. - Higher cost.
Mining 	- Shallow mining. - Lower strip ratio. - Higher productivity. - No blasting required. - Lower cost.	- Higher strip ratio. - Lower productivity. - Blasting required. - Higher cost.
Processing 	- Simple process flow sheet. - No comminution (crushing or milling). - Lower capital and operating costs.	- Complex process flow sheet. - Requires comminution and beneficiation. - Higher capital and operating costs.
Environmental 	- Low levels of radionuclides. - Non-radioactive waste. - Progressive rehabilitation of mining footprint.	Possible deleterious elements in waste.

⁴ Argus, "Rare Earth Analytics", Report, April 2022.

FORTHCOMING ANNOUNCEMENTS

January 2023: December 2022 Quarterly Activities Report.

February 2023: Presentation – Bell Potter Unearthed 2023 Virtual Conference.

February – March 2023: Gold assays from drilling at St Anne's, Murchison Gold Project.

February – March 2023: Gold assays from Circle Valley (Anomaly A) extensional drilling.

February – March 2023: Rare earth assays from Circle Valley infill drilling.

March 2023: Interim Financial Report – half year to 31 December 2022.

March 2023: Initial Mineral Resource – St Anne's, Murchison Gold Project.

April 2023: March 2023 Quarterly Activities Report.

May 2023: Presentation – RIU Sydney Resources Round-up Conference.

June 2023: Initial Mineral Resource – Circle Valley (rare earths).

June 2023: Pre-feasibility Study for the Murchison Gold Project.

July 2023: June 2023 Quarterly Activities Report.

August 2023: Presentation – Australian Gold Conference, Sydney.

This announcement has been authorised for release by the Company's Board of Directors.

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ABOUT MEEKA

Meeka Metals Limited is a gold and rare earths company with a portfolio of high quality 100% owned projects across Western Australia.

Gold

Meeka's flagship Murchison Gold Project has a combined 343km² landholding in the prolific Murchison Gold Fields and hosts a large high-grade 1.2Moz JORC Resource. The Company is actively growing these Resources while also progressing toward production. The release of the Murchison Gold Project Scoping Study in December 2021 outlined a robust Project that produces over 420koz of gold⁵.

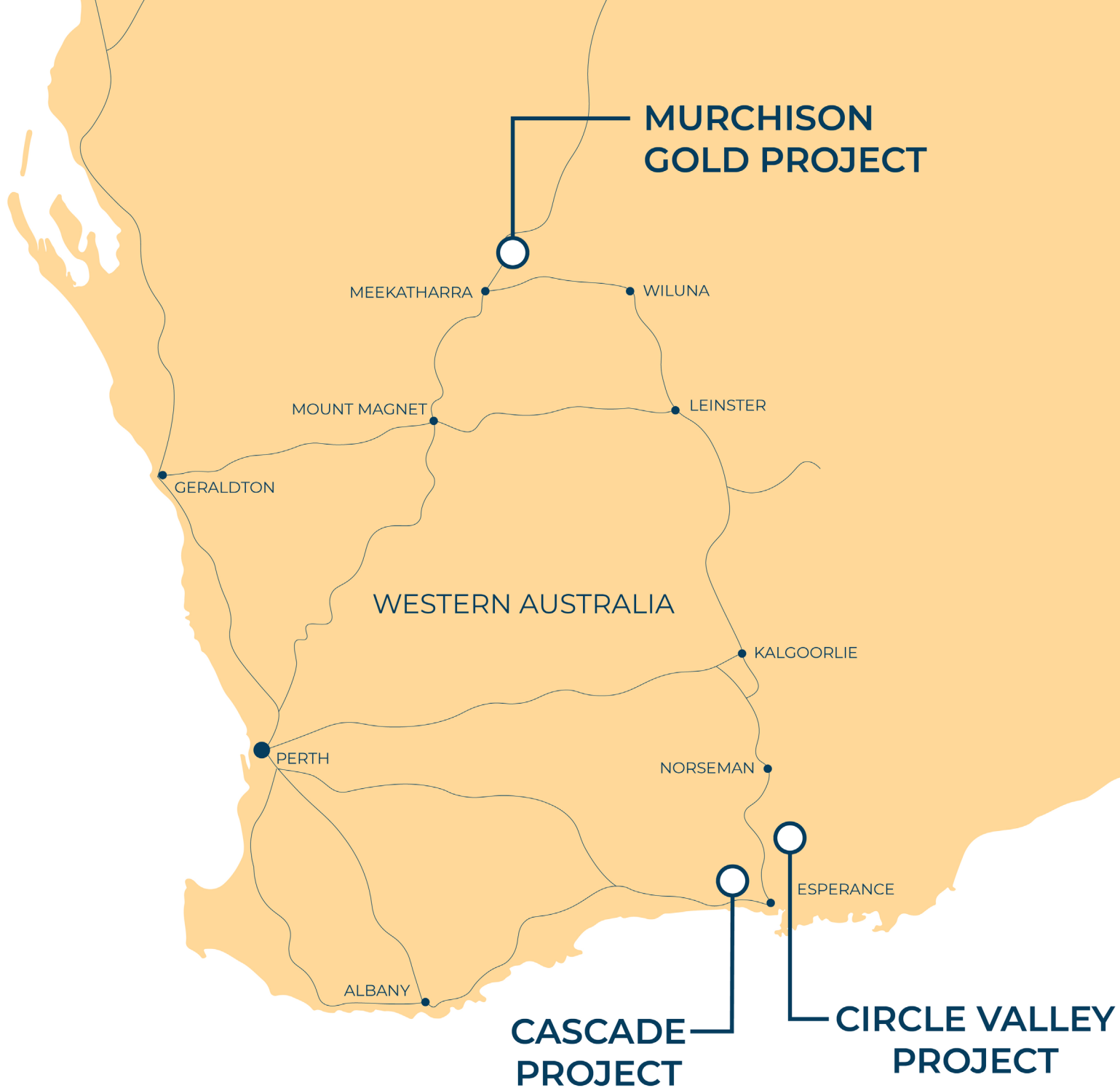
In addition, Meeka owns the Circle Valley Project (222km²) in the Albany-Fraser Mobile Belt (also host to the Tropicana gold mine – 3Moz past production). Gold mineralisation has been identified in four separate locations at Circle Valley and presents an exciting growth opportunity, which is being aggressively pursued.

Rare Earths

Meeka controls the Cascade Rare Earths Project (2,269km²) in a region that is rapidly emerging as a highly prospective clay rare earths province. Importantly, the results to date contain high levels of permanent magnet metals (neodymium-praseodymium). These metals are geopolitically critical, and the Company intends to accelerate its understanding of Cascade through metallurgical work and ongoing drilling.

Circle Valley also hosts clay rare earths within thick, near surface mineralised zones below shallow transported cover. The mineralisation consistently demonstrates a high proportion of the grade as neodymium-praseodymium oxides. Metallurgical work, in addition to infill and extensional drilling are ongoing. An initial Mineral Resource is targeted for 2023.

⁵ First reported in ASX announcement dated 1 December 2021 and titled "Murchison Gold Project Scoping Study". The company confirms that it is not aware of any new information or data that materially affects the information included in the original announcement and that all material assumptions and technical parameters underpinning the target continue to apply and have not materially changed. The company confirms that the form and context in which the Competent Person's findings are presented have not been materially modified from the original market announcement.



Mineral Resource Summary

Project	Measured			Indicated			Inferred			Total		
	Tonnes (¹ 000t)	Grade (g/t)	Ounces (¹ 000oz)	Tonnes (¹ 000t)	Grade (g/t)	Ounces (¹ 000oz)	Tonnes (¹ 000t)	Grade (g/t)	Ounces (¹ 000oz)	Tonnes (¹ 000t)	Grade (g/t)	Ounces (¹ 000oz)
Andy Well	150	11.4	55	1,050	9.3	315	650	6.5	135	1,800	8.6	505
Turnberry				4,600	1.6	230	6,000	2.4	455	10,600	2.0	685
TOTAL	150	11.4	55	5,650	3.0	545	6,650	2.8	590	12,400	3.0	1,190

Notes:

1. Mineral Resources reported to the ASX on 3 January 2023. The company confirms that it is not aware of any new information or data that materially affects the information included in the original announcement and that all material assumptions and technical parameters underpinning the estimates continue to apply and have not materially changed. The company confirms that the form and context in which the Competent Person's findings are presented have not been materially modified from the original market announcement.
2. Mineral Resources are classified in accordance with JORC code (2012).
3. Andy Well Mineral Resource is reported using 0.1g/t cut-off grade.
4. The Turnberry open pit Mineral Resource is only the portion of the Mineral Resource that is constrained within a A\$2,600/oz optimised pit shell and above a 0.5g/t gold cut-off grade.
5. The Turnberry underground Mineral Resource is only the portion of the Mineral Resource that is located outside the A\$2,600/oz optimised pit shell and above a 1.5g/t gold cut-off grade.
6. Estimates are rounded to reflect the level of confidence in the Mineral Resources at the time of reporting.

COMPETENT PERSON'S STATEMENT

The information that relates to Exploration Results as those terms are defined in the 2012 Edition of the "Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserve", is based on information reviewed by Mr Duncan Franey, a Competent Person who is a member of The Australasian Institute of Mining and Metallurgy and the Australian Institute of Geoscientists. Mr Franey is a full-time employee of the Company. Mr Franey has sufficient experience that is relevant to the style of mineralisation and type of deposit under consideration and to the activity being undertaken to qualify as a Competent Person as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves'. Mr Franey consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

The information that relates to Mineral Resources for Andy Well was first reported by the Company in its announcement to the ASX on 18 May 2021. The Company is not aware of any new information or data that materially affects the information included in this announcement and that all material assumptions and technical parameters underpinning the estimates continue to apply and have not materially changed.

The information in this report that relates to Mineral Resources for the Turnberry deposit is based upon information compiled by Mr Mark Roux, a Competent Person who is a member of The Australasian Institute of Mining and Metallurgy. Mr Roux is an independent consultant to the Company. Mr Roux has sufficient experience that is relevant to the style of mineralisation and type of deposit under consideration and to the activity being undertaken to qualify as a Competent Person as defined in the 2012 Edition of the 'Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves'. Mr Roux consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

The information that relates to Scoping Study results is based on information compiled by Mr Tim Davidson, a Competent Person who is a Member of the Australian Institute of Mining and Metallurgy. Mr Davidson is a full-time employee of the company. Mr Davidson is eligible to participate in short and long-term incentive plans of and holds shares and performance rights in the Company as previously disclosed. Mr Davidson has sufficient experience in the study, development and operation of gold projects and consents to the inclusion in the report of the matters based on his information in the form and context in which it appears.

FORWARD LOOKING STATEMENTS

Certain statements in this report relate to the future, including forward looking statements relating to the Company's financial position, strategy and expected operating results. These forward-looking statements involve known and unknown risks, uncertainties, assumptions and other important factors that could cause the actual results, performance or achievements of the Company to be materially different from future results, performance or achievements expressed or implied by such statements. Actual events or results may differ materially from the events or results expressed or implied in any forward-looking statement and deviations are both normal and to be expected. Other than required by law, neither the Company, their officers nor any other person gives any representation, assurance or guarantee that the occurrence of the events expressed or implied in any forward-looking statements will actually occur. You are cautioned not to place undue reliance on those statements.